Omtenta HT24 SF2940

1.) a) X_n converges in distribution to X, as $n \to \infty$,

if $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$, for all continuity points x of the CDF F_X. Here F_{X_n} are the CDFs of the X_n's.

b.) By definition:

$$\varphi_{X_n}(t) = \mathbb{E}\left[e^{itX_n}\right], \quad t \in \mathbb{R}$$

$$= \sum_{k=0}^{n-1} e^{itk}\, \mathbb{P}(X_n = k)$$

$$= \frac{1}{n} \sum_{k=0}^{n-1} e^{itk}$$

hint / finite geometric sum

$$= \frac{1}{n} \frac{1 - e^{itn}}{1 - e^{it}}.$$

Answer: $\varphi_{X_n}(t) = \frac{1}{n} \frac{1 - e^{itn}}{1 - e^{it}}$.

c.) The characteristic function of Y_n is

$$\varphi_{Y_n}(t) = \varphi_{X_n}\left(\frac{t}{n}\right) = \frac{1}{n} \frac{1 - e^{it}}{1 - e^{\frac{it}{n}}}.$$

For large n, we have $1 - e^{\frac{it}{n}} = 1 - \left(1 + \frac{it}{n} + o\left(\frac{1}{n}\right)\right)$

$$= -\frac{it}{n} + o\left(\frac{1}{n}\right).$$

Thus, $\varphi_{Y_n}(t) = \frac{1 - e^{it}}{-it + o(1)} \xrightarrow{n \to \infty} \frac{e^{it} - 1}{it}$.

This we recognize as the characteristic function of a random variable $Y \in U(0,1)$.

Answer: $Y_n \xrightarrow[n\to\infty]{d} Y$ with $Y \in U(c,1)$ by the Uniqueness theorem.

2.) Assuming $\vec{X}$ is a bivariate (non-degenerate) Gaussian, its density is

$$f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi\sqrt{\det\Sigma}}\, e^{-\frac{1}{2}(x_1,x_2)\Sigma^{-1}\begin{pmatrix}x_1\\x_2\end{pmatrix}}.$$

Indeed, the given density has no linear term in the exponential function, hence $\vec{\mu}_{\vec{X}} = \begin{pmatrix}0\\0\end{pmatrix}$.

Comparing we see that

$$\frac{1}{2}(x_1,x_2)\Sigma^{-1}\begin{pmatrix}x_1\\x_2\end{pmatrix} = x_1^2 - x_1\cdot x_2 + x_2^2$$

with $\Sigma^{-1} = \begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}$, $\det(\Sigma^{-1}) = 3$.

Using Cramer's rule, we find the inverse of Σ^{-1}:

$$\Sigma = \frac{1}{\det(\Sigma^{-1})}\cdot\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix} = \frac{1}{3}\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}.$$

Hence we can write:

$$f_{X_1,X_2}(x_1,x_2) = \frac{\sqrt{3}}{2\pi} e^{-\frac{1}{2}(x_1,x_2)\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}.$$

Thus indeed $\vec{X}$ is a bivariate Gaussian and

$$\vec{X} \in \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}\right). \quad //$$

b.) From a) it is immediate that $c = \frac{\sqrt{3}}{2\pi}$, when we used that $\det \Sigma = \frac{1}{3}$. //

c.) Following the hint, we consider the moment generating function $\Psi_{X_1,X_2}(t_1,t_2)$

$$\Psi_{X_1,X_2}(t_1,t_2) = e^{\frac{1}{2}(t_1,t_2)\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}\begin{pmatrix} t_1 \\ t_2 \end{pmatrix}}$$

$$= e^{\frac{1}{3}(t_1^2 + t_1 \cdot t_2 + t_2^2)}.$$

We obtain the moments by computing derivatives of ψ_{X_1,X_2} at $\begin{pmatrix} t_1 \\ t_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$: $\mathbb{E}[X_1^2 X_2] = \frac{\partial}{\partial t_2} \frac{\partial^2}{\partial t_1^2} \psi_{X_1,X_2}(t_1,t_2)\Big|_{t_1=t_2=0}$.

We now compute,

$$\frac{\partial}{\partial t_1} \psi_{X_1,X_2}(t_1,t_2) = \frac{1}{3}(2t_1+t_2)\,\psi_{X_1,X_2}(t_1,t_2)$$

$$\frac{\partial^2}{\partial t_1^2} \psi_{X_1,X_2}(t_1,t_2) = \frac{2}{3}\psi_{X_1,X_2}(t_1,t_2) + \frac{1}{9}(2t_1+t_2)^2\,\psi_{X_1,X_2}(t_1,t_2).$$

Finally,

$$\frac{\partial}{\partial t_2}\frac{\partial^2}{\partial t_1^2} \psi_{X_1,X_2}(t_1,t_2) = \frac{2}{9}(2t_2+t_1)\,\psi_{X_1,X_2}(t_1,t_2) + \frac{2}{9}(2t_1+t_2)\,\psi_{X_1,X_2}(t_1,t_2) + \frac{1}{9}(2t_1+t_1)^2 \cdot \frac{1}{3}(2t_2+t_1)\,\psi_{X_1,X_2}(t_1,t_2)$$

Setting $t_1=t_2=0$, we obtain

$$\frac{\partial}{\partial t_2}\frac{\partial^2}{\partial t_1^2}\psi_{X_1,X_2}(t_1,t_2)\Big|_{t_1=t_2=0} = C\cdot\underbrace{\psi_{X_1,X_2}(0,0)}_{=1} = 0.$$

Answer: $\mathbb{E}[X_1^2 X_2] = 0$.

Alternative to c.): We aim to compute

$$\mathbb{E}[X_1^2\cdot X_2] = C\int_{\mathbb{R}}\int_{\mathbb{R}} x_1^2 x_2\cdot e^{-x_1^2+x_1\cdot x_2-x_2^2}\,dx_1\,dx_2, \quad C=\frac{\sqrt{3}}{2\pi}.$$

If we change the variables $\begin{cases} y_1 = -x_1 \\ y_2 = -x_2 \end{cases}$ then we get

$$\mathbb{E}[X_1^2\cdot X_2] = C\int_{\mathbb{R}}\cdot\int_{\mathbb{R}} y_1^2(-y_2)\,e^{-y_1^2+y_1y_2-y_2^2}\,dy_1\,dy_2,$$

(since the Jacobian is 1)

$$= -C\int_{\mathbb{R}}\int_{\mathbb{R}} y_1^2 y_2\, e^{-y_1^2+y_1y_2-y_2^2}\,dy_1\,dy_2$$

$$= -\mathbb{E}[X_1^2\cdot X_2].$$

Hence $\mathbb{E}[X_1^2 X_2]$ must be zero. //

3.) Since the X_i's are i.i.d., are centered and have finite variance, we have the CCT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \xrightarrow[n\to\infty]{d} \mathcal{N}(0, \sigma^2),$$

where we used that $\mathbb{E} X_i = 0$. Since $X_i \in U(-1,1)$, we have

$$\sigma^2 = \frac{1}{2}\int_{-1}^{1} x^2 \, dx = \frac{1}{6}\left(1^3 - (-1)^3\right) = \frac{1}{3}.$$

Next, we look at the CDF of U_n:

$$\mathbb{P}(U_n \le x) = \mathbb{P}(X_1 \le x, X_2 \le x, \dots, X_n \le x)$$

i.i.d.

$$= \mathbb{P}(X_1 \le x)^n.$$

Now, $\mathbb{P}(X_1 \le x) = \begin{cases} 0, & x \le -1, \\ \frac{1+x}{2}, & x \in (-1,1), \\ 1, & x \ge 1. \end{cases}$

Consider next, $x \in (0,1)$, then

$$\mathbb{P}(X_1 \le x)^n = \left(\frac{1+x}{2}\right)^n \xrightarrow[n\to\infty]{} 0,$$

because $\left|\frac{1+x}{2}\right| < 1$. Thus $\mathbb{P}(U_n \le x) \xrightarrow[n\to\infty]{} \begin{cases} 0, & x < 1, \\ 1, & x \ge 1. \end{cases}$

In other words, U_n converges in distribution to the constant one.

Therefore, $U_n \xrightarrow[n\to\infty]{P} 1$.

Similarly:

$$\mathbb{P}(V_n \le x) = 1 - \mathbb{P}(V_n > x)$$

$$= 1 - \mathbb{P}(X_1 > x, X_2 > x, \dots, X_n > x)$$

$$\overset{\text{i.i.d.}}{=} 1 - \mathbb{P}(X_1 > x)^n.$$

Let now $x \in (-1,1)$, then

$$\mathbb{P}(V_n \le x) = 1 - \mathbb{P}(X_1 > x)^n$$

$$= 1 - \left(\frac{1-x}{2}\right)^n \xrightarrow[n\to\infty]{} 1.$$

On the other hand, $\mathbb{P}(V_n \le x) = 0$ for $x < -1$, and $\mathbb{P}(V_n \le x) = 1$, for $x > 1$.

Thus $V_n \xrightarrow{d} -1$ as $n \to \infty$, and we conclude that

$$V_n \xrightarrow{P} -1 \text{ as } n \to \infty.$$

Next, we use Cramér-Slutsky:

$$\underbrace{\frac{1}{\sqrt{n}}}_{\xrightarrow{d} \mathcal{N}(0,1/3) \text{ as } n\to\infty} S_n \cdot \underbrace{\frac{U_n}{V_n}}_{\xrightarrow{P} -1} \xrightarrow[n\to\infty]{d} -\mathcal{N}(0,1/3).$$

($U_n \xrightarrow{P} 1$)

Since $\mathcal{N}(0,1/3)$ is symmetry $-\mathcal{N}(0,1/3) \overset{d}{=} \mathcal{N}(0,1/3)$,

and we conclude:

Answer: $$\frac{1}{\sqrt{n}} \frac{S_n \cdot U_n}{V_n} \xrightarrow[n\to\infty]{d} \mathcal{N}(0,1/3). \quad //$$

4.) a. By definition $f_{\bar{X}|\bar{Y}=y}(x) = \dfrac{f_{\bar{X},\bar{Y}}(x,y)}{f_{\bar{Y}}(y)}$

mixing integral

$$= \frac{f_{X,Y}(x,y)}{\int_{\mathbb{R}} f_{Y|X=x}(y)\,dy}.$$

On the other hand, we have

$$f_{\bar{Y}|\bar{X}=x}(y) = \frac{f_{\bar{X},\bar{Y}}(x,y)}{f_{\bar{X}}(x)}.$$

Combining the above, we get Bayes formula

$$f_{\bar{X}|\bar{Y}=y}(x) = \frac{f_{\bar{Y}|\bar{X}=x}(y)\, f_{\bar{X}}(x)}{\int_{\mathbb{R}} f_{\bar{Y}|\bar{X}=\hat{x}}(y) f_{\bar{X}}(\hat{x})\,d\hat{x}}.$$

b.) We recognize the broken stick from the book and lectures:

$$\mathbb{E}[Y|X=x] = \int_{\mathbb{R}} y \underbrace{f_{Y|X=x}(y)}_{\frac{1}{x}\mathbb{1}_{\{0<y<x\}}} dy, \quad x \in (0,1).$$

$$= \int_0^x y \frac{1}{x} dy = \frac{x}{2}.$$

The conditional expectation is thus

$$\mathbb{E}[Y|X] = \frac{X}{2}. \qquad //$$

c.) We can use the mixing integral,

$$f_Y(y) = \int_{\mathbb{R}} \underbrace{f_{Y|X=x}(y)}_{\frac{1}{x}\mathbb{1}_{\{0<y<x\}}} \underbrace{f_X(x)}_{1\cdot\mathbb{1}_{\{0<x<1\}}} dx, \quad y \in (0,1).$$

$$= \int_y^1 \frac{1}{x} dx = \underbrace{\ln(1)}_{=0} - \ln(y) = -\ln(y).$$

Answer: $f_Y(x) = \begin{cases} -\ln(y), & y \in (0,1) \\ 0, & \text{else} \end{cases}$ //

d.) By definition, we have

$$\mathbb{E}[X \mid Y = y] = \int_{\mathbb{R}} x \cdot f_{X|Y=y}(x)\, dx$$

Now Bayes formula yields

$$f_{X|Y=y}(x) = \frac{f_{Y|X=x}(y) \cdot f_X(x)}{\int f_{Y|X=x}(y) f_X(x)\, dx}, \quad x \in (0,1)$$

($\frac{1}{x} \mathbb{1}_{\{0<y<x<1\}}$; denominator computed in c.)

Thus,

$$\mathbb{E}[X \mid Y = y] = \frac{\int_y^1 x \cdot \frac{1}{x} \mathbb{1}_{\{0<y<x<1\}}\, dx}{-\ln y}$$

$$= \frac{1-y}{-\ln(y)} \quad \Uparrow \text{ so } c_y < 13.$$

The conditional expectation is thus given by

$$\boxed{E[X|Y] = \frac{1-\bar{Y}}{-\ln(\bar{Y})}}.$$

//